MINING DEALER BEST PRACTICE

All Type Gear Inspection Stand

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

March 2021
4.03-210316-1

1.0 Introduction

Gears and gear systems are used for power transmissions in various components. Since these gears are under intense operating conditions, the parts must be inspected with great care and attention. If damaged gears are not found, the rebuilt component may fail when it's put back into service.

To inspect the gears more effectively, controlling them with a stand under appropriate lighting will produce more positive results.

2.0 Best Practice Description

A gear inspection stand will provide better accessibility to and more accurate results of all sizes of gears in a component. All gears from a single component can be inspected at the same time with this stand. With the light attached to the stand, thorough inspection of the gears can be achieved and even minimal damage and wear can be found.

Steel tubes, a weldable steel profile and an LED lamp are used in this project. The engineering prints for the stand can be found in the attachments section. Below are pictures of the manufactured stand.

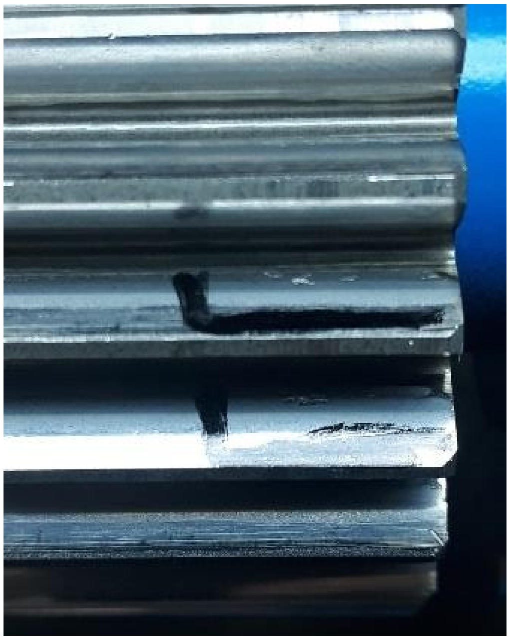
3.0 Implementation Steps

1. Different types of gears are placed on the appropriate control rod.

2. The moving lamp is brought closer to the gears to be inspected and the light height is adjusted.

3. The stand is controlled from both sides by turning the gears under the light.

4. Inspection of the gears for any damage or wear is performed.



4.0 Benefits

- Easy inspection of all types of gears at the same time;
- Produces more accurate results of inspections;
- Minimizes overlooking gear damage, such as spot fractures on the gear planetary teeth.

5.0 Resources Required

Item	Cost
Material	$170
Labor	$390
Total	$560

6.0 Supporting Attachments / References

All Type Gear Inspection Stand_Engineering Print.pdf

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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